Sham Krishna Majumder

M.S. Candidate in Physics · University of Dhaka, Bangladesh
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Research Interests

My research centres on **statistical physics and complex systems**, with a focus on percolation theory, critical phenomena, non-equilibrium statistical mechanics, and disordered systems on complex spatial networks. I am also interested in condensed matter physics, biophysics, and non-linear dynamics.

Methods: Monte Carlo simulation (Newman–Ziff algorithm), finite-size scaling and data collapse, critical exponent and threshold analysis, high-performance computing (C++/OpenMP).

Education

Master of Science in Physics May 2025 – Expected Dec 2026
University of Dhaka *Dhaka, Bangladesh*

- **Research area:** Non-equilibrium Statistical Mechanics, Percolation Theory, Weighted Planar Stochastic Lattices
- **Advisor:** Dr. Md. Kamrul Hassan, Department of Physics, University of Dhaka

Bachelor of Science in Physics Jan 2020 – May 2025
University of Dhaka *Dhaka, Bangladesh*

- **CGPA:** 3.41 / 4.00

Research Experience

Critical Phenomena and Phase Transitions on Weighted Planar Stochastic Lattices (WPSL) May 2025 – Present
Supervisor: Dr. Md. Kamrul Hassan, University of Dhaka

- Study site and bond percolation on WPSL and its variants, a disordered planar lattice, analysing critical thresholds, order-parameter dynamics, and scaling behaviour to determine universal exponent sets.
- Developed parallel C++ simulation code (OpenMP) on Linux servers to compute percolation observables, order parameters, Shannon entropy, and Binder cumulants across system sizes.
- Applied finite-size scaling and data collapse to extract critical exponents and verify scaling relations and universality.
- Source code: `github.com/Skrishna36`

Selected Academic Projects

- Monte Carlo Simulations of Random-Walk Variations
- Construction and Degree Distribution of the Weighted Planar Stochastic Lattice, WPSL1
- Lattice, Degree Distribution and Spanning Probability of WPSL2
- Percolation on the Square Lattice

Fellowships and Awards

National Science and Technology (NST) Fellowship 2025
Ministry of Science and Technology, Government of Bangladesh

- Awarded for M.S. thesis research in Physics.

Technical Skills

Programming	C++, Python (NumPy, Matplotlib), R, MATLAB, SQL
Tools	Linux/Bash, Git, Gnuplot, LaTeX, remote HPC workflows
Computational methods	Monte Carlo simulation, OpenMP parallelisation, finite-size scaling
Languages	Bengali (native); English (proficient, IELTS 7.5)

Teaching and Mentoring

Programming Instructor 2024 – 2025
Physics Department Career Club, University of Dhaka *Dhaka, Bangladesh*

- Designed and taught short courses on introductory and intermediate C++ programming for physics students.
- Covered object-oriented programming, memory management, and algorithmic problem solving; mentored students in writing clean code for scientific applications.

Physics Instructor and Academic Tutor (2022 – 2025): taught HSC-level physics, including mechanics, electromagnetism, thermodynamics, and modern physics, to students preparing for university admission examinations.

Hobbies and Creative Interests

I create comics, draw digitally, and enjoy reading and writing short stories.

References

Dr. Md. Kamrul Hassan
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